

Ex-4 Scenario-Based Report Development Using Prompt Engineering Techniques

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Subject: 19TD608-Prompt Engineering

Slot: T2-H16

Aim

To develop a scenario-based report using advanced prompt engineering techniques and evaluate outputs from multiple AI tools.

Scenario Chosen

AI in Education

Artificial Intelligence is transforming education through personalized learning, automated assessments, virtual tutors, and smart content generation. This experiment compares how different AI tools respond to various prompting techniques related to AI in education.

AI Tools Used

1. [ChatGPT](#)

2. [Google Gemini](#)

Algorithm

1. Select a scenario related to AI applications.
2. Create prompts using different prompting techniques.
3. Execute the same prompts in ChatGPT and Gemini.
4. Record outputs from both AI tools.
5. Compare outputs based on:
 - Accuracy
 - Clarity
 - Completeness
 - Response Quality
 - Ease of Understanding
6. Analyze differences between tools.
7. Prepare the report with screenshots and comparison tables.
8. Convert the document into PDF format and submit.

Prompting Techniques and Comparison

1. Comparative Analysis Prompt

Prompt

“Compare AI in education and healthcare.”

ChatGPT Output

AI in education helps personalize learning, automate grading, and provide virtual tutoring. AI in healthcare assists in disease prediction, medical imaging, and patient monitoring. Education focuses on improving learning experiences, while healthcare focuses on improving diagnosis and treatment.

Gemini Output

AI in education is like a super-smart **assistant teacher**, helping students learn at their own pace and grading quizzes. If it makes a mistake, it's easily fixed.

AI in healthcare is like a **doctor's co-pilot**, spotting diseases in X-rays and predicting risks. Because mistakes here can be dangerous, it faces incredibly strict safety rules.

Ultimately, both fields use AI to cut through boring paperwork so humans can focus on what matters: teaching students and caring for patients.

Comparison

Criteria	ChatGPT Output	Gemini Output
Writing Style	Formal and structured	Conversational and engaging
Clarity	Very clear and direct	Very easy to understand with examples
Explanation Method	Point-by-point comparison	Uses real-life analogies
Technical Depth	Moderate	Moderate to High
Creativity	Basic explanatory style	Creative explanation style
Ease of Understanding	Good for academic use	Excellent for beginners
Focus	Educational and healthcare functions	Human-centered applications and safety

Special Highlight	Mentions personalized learning and medical diagnosis	Explains AI as “assistant teacher” and “doctor’s co-pilot”
Response Quality	Structured and concise	More interactive and relatable
Overall Impression	Best for reports and documentation	Best for simple understanding and engagement

2. Experiential Perspective Prompt

Prompt

“Explain AI usage from a teacher’s perspective.”

ChatGPT Output

As a teacher, AI helps create lesson plans, generate quizzes, track student performance, and provide personalized support for students. It reduces workload and improves classroom management.

Gemini Output

For a teacher, AI is like a tireless assistant that handles the exhausting paperwork. It instantly drafts lesson plans, creates quizzes, and customizes reading materials for different student levels in seconds, saving hours of prep time. While it serves as a great creative partner for fresh classroom ideas, it also adds a stressful new duty: forcing teachers to play detective to make sure students aren’t using chatbots to cheat on their homework.

Comparison

Criteria	ChatGPT Output	Gemini Output
Writing Style	Simple and professional	Conversational and descriptive
Perspective Used	Teacher's daily teaching support	Teacher's real-life experience and challenges
Clarity	Very clear and direct	Clear with detailed explanation
Creativity	Moderate	High
Completeness	Covers lesson plans, quizzes, and student support	Covers preparation help, creativity, and cheating concerns
Emotional/Human Touch	Limited	Strong human perspective
Technical Detail	Basic	More detailed
Ease of Understanding	Excellent	Excellent
Special Highlight	Focuses on reducing workload	Highlights both advantages and challenges of AI

**Response
Quality**

Structured and concise

Realistic and engaging

**Overall
Impression**

**Good for academic
reporting**

**Better for experiential
understanding**

3. Everyday Functioning Prompt

Prompt

“Explain how AI helps in daily life.”

ChatGPT Output

AI helps people through voice assistants, navigation apps, online shopping recommendations, translation tools, and smart devices used at home.

Gemini Output

In daily life, AI acts as an invisible assistant that quietly streamlines your day behind the scenes. It maps out the fastest driving route by analyzing live traffic, filters spam out of your inbox, and curates your entertainment by predicting which songs or shows you will enjoy next. From unlocking your phone with facial recognition to predicting the next word you want to type, AI eliminates minor daily frictions so smoothly that you rarely even notice it is working.

Comparison

Criteria

ChatGPT Output

Gemini Output

Writing Style

Simple and straightforward

Descriptive and narrative

Explanation Approach	Direct explanation with examples	Real-life storytelling approach
Clarity	Very clear	Very clear and engaging
Creativity	Moderate	High
Completeness	Covers common AI applications	Covers detailed day-to-day AI interactions
Technical Depth	Basic	Moderate
Ease of Understanding	Excellent	Excellent
User Engagement	Informative	Highly relatable
Special Highlight	Focuses on AI tools like assistants and smart devices	Emphasizes AI as an “invisible assistant” working in the background
Response Quality	Concise and structured	Detailed and immersive
Overall Impression	Suitable for academic summaries	Better for practical real-world understanding

4. Universal Prompt

Prompt

“Explain AI for beginners.”

ChatGPT Output

Artificial Intelligence is a technology that allows machines to think, learn, and solve problems like humans. Examples include chatbots, recommendation systems, and virtual assistants.

Gemini Output

Artificial Intelligence (AI) is essentially training computers to think and learn from data, rather than just following rigid rules. By analyzing massive amounts of information, AI recognizes patterns to solve problems, automate tasks, and make predictions. It is the hidden engine behind everyday tech, from your phone's autocorrect and streaming recommendations to virtual assistants and navigation apps, constantly improving the more data it processes.

Comparison

Criteria	ChatGPT Output	Gemini Output
Writing Style	Simple and educational	Conversational and descriptive
Clarity	Very clear	Very clear and relatable
Explanation Method	Definition-based explanation	Real-life example-based explanation
Creativity	Moderate	High
Completeness	Basic overview of AI	Detailed practical understanding
Ease of Understanding	Excellent for beginners	Excellent for all users
Human Connection	Limited	Strong practical connection
Response Quality	Structured and concise	Engaging and immersive

5. Prompt Structure Refinement

Prompt

“Explain AI with examples, advantages, and limitations.”

ChatGPT Output

AI is the simulation of human intelligence in machines.

Examples include chatbots, self-driving cars, and recommendation systems.

Advantages

- Saves time
- Improves accuracy
- Automates repetitive tasks

Limitations

- Expensive implementation
- Requires large data
- May reduce human jobs

Gemini Output

Artificial Intelligence (AI) simulates human intelligence by allowing computers to learn from massive amounts of data, recognize patterns, and make independent decisions.

Advantages

- **Works 24/7:** AI never gets tired, handles customer support around the clock, and stays accurate without needing breaks.
- **Lightning Fast:** It analyzes huge piles of data in seconds, finding patterns that would take humans months to spot.
- **Handles Boring Tasks:** By taking over repetitive jobs like sorting files or scheduling, it frees humans up for creative work.

Limitations

- **Can Be Biased:** AI learns from human data. If that data is unfair or flawed, the AI will make unfair decisions too.
- **No Real Common Sense:** It matches patterns but doesn't actually understand anything. It has zero empathy or true human instinct.
- **Costs a Lot:** Building and running smart AI systems takes massive amounts of money, tech skills, and electrical power.

Criteria	ChatGPT Output	Gemini Output
Writing Style	Simple and structured	Detailed and explanatory
Organization	Clearly divided into examples, advantages, and limitations	Well-structured with detailed points
Clarity	Very clear and concise	Clear with deeper explanation
Creativity	Moderate	High
Completeness	Covers basic points	Covers broader and realistic points
Technical Depth	Basic to moderate	Moderate to high
Ease of Understanding	Excellent	Excellent
Human Insight	Limited	Strong practical insight
Special Highlight	Focuses on common AI applications	Explains bias, lack of common sense, and operational cost

Response
Quality

Academic and concise

Analytical and detailed

6. Prompt Size Limitation

Prompt

“Explain AI in 3 lines.”

ChatGPT Output

AI enables machines to learn and make decisions.
It is used in healthcare, education, and business.
AI improves automation and efficiency.

Gemini Output

Artificial Intelligence (AI) is a technology that trains computers to think, learn, and solve problems by finding patterns in massive amounts of data. It powers everyday tools like your phone's smart recommendations and navigation apps to make life easier by automating complex or repetitive tasks. While it mimics human decision-making at lightning speed, it lacks actual common sense and relies completely on the information we give it.

Comparison

Criteria	ChatGPT Output	Gemini Output
Writing Style	Short and direct	Detailed and descriptive
Clarity	Very simple	Clear and informative
Response Length	Very concise	Slightly longer
Creativity	Moderate	High
Completeness	Covers basic AI uses	Explains AI working and limitations

Ease of Understanding	Excellent	Excellent
Response Quality	Compact and academic	Informative and engaging
Overall Impression	Best for strict word limits	Best for meaningful concise explanation

OVERALL COMPARISON OF CHATGPT VS GEMINI (ACROSS 6 PROMPTING TECHNIQUES)

Evaluation Criteria	ChatGPT	Gemini	Better Tool
Accuracy	Excellent (9/10) Provides accurate and reliable information.	Excellent (9/10) Provides accurate and reliable information.	Equal
Clarity	Excellent (10/10) Very clear and easy to understand.	Very Good (9/10) Clear with good explanations.	ChatGPT
Completeness	Good (8/10) Covers all key points but in a limited depth.	Excellent (10/10) More complete with detailed explanations.	Gemini
Creativity	Moderate (7/10) Basic and straightforward responses.	High (10/10) Creative and engaging explanations.	Gemini
Technical Detail	Moderate (7/10) Explains concepts in simple terms.	High (9/10) Includes more technical and practical details.	Gemini
Ease of Understanding	Excellent (10/10) Very easy to understand for beginners.	Excellent (10/10) Easy to understand with real-life examples.	Equal
Human-Like Explanation	Moderate (7/10) Less human-like in tone.	Excellent (10/10) More human-like and relatable.	Gemini
Structured Formatting	Excellent (10/10) Well-structured and organized.	Very Good (9/10) Well-structured but slightly less formal.	ChatGPT
Beginner Friendliness	Excellent (10/10) Best for beginners and students.	Very Good (9/10) Beginner-friendly with examples.	ChatGPT
Real-World Relatability	Good (8/10) Provides real-world examples.	Excellent (10/10) Highly relatable and practical examples.	Gemini
Overall Conclusion	Best for concise, structured, and academic explanations.	Best for detailed, creative, and real-world understanding.	Both are excellent in their own strengths.

Findings

- ChatGPT provided more user-friendly and simplified explanations.
- Gemini generated more technical and detailed responses.
- Both AI tools produced accurate outputs for all prompting techniques.
- Prompt refinement improved the quality and structure of responses.
- Short prompts generated concise answers but reduced detail.

Result

Thus, scenario-based prompt engineering techniques were successfully applied using multiple AI tools. The outputs were compared and evaluated based on clarity, accuracy, completeness, and response quality.

References

1. [ChatGPT by OpenAI](#)
2. [Google Gemini](#)
3. [OpenAI Official Website](#)
4. [Google AI Official Website](#)